1.FACTORIAL WITH RECURSIVE METHOD:

Recursion is a technique in which a function calls itself repeatedly to solve a problem until it reaches a base condition.

ALGORITHM:

1. Start.
2. Read a number n from the user.
3. Call the factorial function with n.
4. Check if n <= 1.
5. If true, return 1.
6. Otherwise, call the function with n - 1.
7. Multiply n with the returned value.
8. Return the result.
9. Display the factorial.
10. STOP.

Time Complexity:

Case	Time Complexity
Best Case	**O(1)**
Average Case	**O(n)**
Worst Case	**O(n)**

Space Complexity:

Space Complexity **= O(n)**

OUTPUT:

```
Enter a number: 5
Factorial = 120

...Program finished with exit code 0
Press ENTER to exit console.
```

Conclusion :

The factorial of a number was successfully calculated using recursion. The function repeatedly calls itself with a smaller value until the base condition is reached. This method is easy to understand for recursive problems, but it uses more memory because of function calls. It has **O(n) time complexity and O(n) space complexity**.